TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Seamless Private cellular and wi-fi IntegratioN for Enterprises (SPINE)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication (√), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Mobile Communication	Mobile Communication Systems	Software	Internet SW
Concept of Proposed Technology	Many enterprises are heavily investing in private cellular (5G+) networks, in a wide range of application domains such as manufacturing, supply chain and shipping, information technology and healthcare. This is driven by the need to have better control over Quality of Service and stability under heavy load. However, private 5G installations and licensing can be expensive and therefore coverage may not be ubiquituous, and the 5G spectrum must be used judiciously to prioritize performance-critical and latency-sensitive applications. Enterprises also have complementary WiFi coverage, which can effectively complement private cellular			

	coverage in such settings. However, switching back-and-forth between 5G and WiFi for mobile devices/users often results in connectivity disruptions and the resulting latencies can be unacceptable for certain applications. In this project, we will develop and test SPINE, a solution for seamlessly integrating private cellular and WiFi for enterprise networks by (a) providing a AI-based solution to map user equipment (UE) to cellular or WiFi based on their precise location, predicted mobility, and latency-sensitivity and (b) using multipath transport protocols to integrate the private cellular technology with WiFi to provide further seamless connectivity and smooth application experience. The solutions will be developed as flexible and open-source software implementations that are compatible with Open-RAN (O-RAN) cellular technologies.
Rationale for Joint R&D Project	We have done the foundational work on integrating cellular and WiFi technologies in laboratory testbeds and demonstrated the benefits of using multipath for seamless connectivity despite disruptions in one or the other path. In this project, these solutions would need to be extended to include UE localization and AI-based mobility prediction and optimization of the handoffs. However, realization of the full commercial potential of this technology will require collaboration with an industry partner on a joint R&D project to develop, deploy, and field test our solutions beyond the small campus networks available in academic institutions. The industry partner will also help the team to wrap the localization, prediction and optimization solutions in a single software package that can be run on a standalone hardware appliance that communicates directly with the 5G and WiFi access points.
Key Words	1. Mobility across multiple access technologies 2. Seamless 5G-WiFi interworking 3. Predictive link layer handoffs 4. Multipath transport protocols

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	This project will develop SPINE, a solution for integrating private cellular and WiFi networks for enterprise communication, providing seamless connectivity and interworking to mobile user and industrial IoT devices, through AI-based UE localization, mobility prediction, handoff optimization, and multipath support. Open-source software implementation of the solution compatible with Open-RAN (O-RAN) cellular technologies, and testing in multiple edge networks in the USA and Korea will validate our solution.			
R&D Target Specifications	We will demonstrate the effectiveness of our approach with two broad types of metrics: (a) connectivity disruption measures such as number of handoffs, and total time taken for handoffs between 5G base stations as well as between 5G and WiFi from the point of view of the mobile device and (b) the average and variance in data rate achieved in several mobility scenarios. The target would be to attain at least 50% improvement in these measures as compared to current practice where cellular or WiFi is chosen reactively based on relative signal strengths only.			
R&D Plan	**Research Approach** This work will be accomplished through a collaboration between researchers at Rensselaer Polytechnic Institute (RPI) and the University of California at Riverside (UCR), in addition to our Korean SME partner(s). RPI will be the lead institution, with funding provided by RPI for the work done at UCR through a sub-contract. **Key tasks** 1. **UE localization:** Wireless signal strengths will be collected from different 5G and WiFi access points (whose locations within the enterprise are known) and analyzed to localize the mobile (user or IoT) device (UE) 2. **Mobility predicton:** Based on the device location and mobility history, AI tools will be used to predict the UE trajectory. 3. **Handoff optimization:** The UE type, location and predicted trajectory (along with the known map of the signal strengths) will then be used by a (potentially dedicated) interworking management engine to manage the seamless interworking between networking technologies, to assign the UE to 5G or WiFi, and perform handoffs between the two. 4. **Multipath support:** If multipath transport technology is available at the UE, the solution will determine whether both technologies should be utilized, or if only one of the available technologies should be used. 5. **Implementation and field testing:** Implement the solution in one software package that will run on a dedicated server (representative of the			

	interworking management engine) end server, and evaluate it in several edge networks served by multiple 5G and WiFi access points. **Methodologies** We will build upon previous as well as our own prior work on UE localization and mobility prediction [CrowdLoc2018], handoff optimization [M&M2024] and multipath transport optimization [Sharma2012]. The main extension to our prior work would be to utilize cutting edge AI tools toward these tasks. The development of the Open-RAN compatible implementation will build upon our experience in developing a 5G-Core emulator [L25GC]. **Milestones and Timelines** *Total project duration:* 42 months (including the initial 6-month go/no-go period) **M1.** Overall design of the solution consisting of the different components and communication protocols [0–6 months, go/no-go milestone] **M2.** Developing and testing the UE localization and mobility prediction solutions using AI tools. [7–15 months] **M3.** Optimizing handoffs between 5G and WiFi [16–24 months] **M4.** Integrating multipath transport protocols [16–24 months] **M5.** Systems integration of solutions into software [25–31 months] **M6.** Integrated testing of prototype in multiple locations in USA (RPI and UCR) and Korea (with the help of Korean SME partner(s)) and evaluating performance metrics. [32–42 months]

4. Role & Responsibility (R&R)

Expected R&R of Research Institute	The US-based research institute comprises two universities (RPI and UCR) led by Dr. Koushik Kar (RPI) and Dr. K. K. Ramakrishnan (UCR), respectively. They will have the following roles and resonsibilities: 1) The overall system design toward milestone M1 will be done jointly between both US universities (RPI and UCR) and Korean SME partner. 2) Developing and testing the UE localization and mobility prediction solutions toward milestone M2 will be done collaboratively between RPI and UCR. 3) Optimization of handoffs between 5G and WiFi toward milestone M3 will be done primarily at RPI. 4) Integration of multipath transport protocols toward milestone M4 will be done collaboratively between RPI and UCR. 5) Systems integration of solutions into a software package toward milestone M5 will be done primarily at UCR in collaboration with Korean SME partner. 6) The field testing of the system toward milestone M6 will be done jointly between both US universities (RPI and UCR) and Korean SME partner.

Expected R&R of Korean SME Partner	The Korean SME partner will have the following roles and resonsibilities: 1) Collaborate with the US institutions in developing the overall system design toward milestone M1. 2) Work with UCR researchers in integrating the solutions into a software package to be run on an edge server, toward milestone M5. 3) Field test the system across locations in Korea to evaluate performance metrics evaluate its commercialization potential, toward milestone M6.

5. Expected Outcome/Impact

Expected Outcomes and Impact	The key outcome from this project will be a software solution that can run on an edge serve and assist the 5G and WiFi access points in the edge network to perform UE handoffs or multipath traffic splitting for seamless connectivity of the UEs, while meeting application specific latency needs. As enterprises continue to invest heavily in private 5G in diverse sectors such as manufacturing and healthcare, SPINE can help them reduce cost in private 5G network installation and maintenance by allowing them to be complemented by WiFi effectively while meeting application needs.

6. References

[Sharma2012] V. Sharma, K. Kar, S. Kalyanaraman and K. K. Ramakrishnan, A Transport Protocol to Exploit Multipath Diversity in Wireless Networks, *IEEE/ACM Transactions on Networking*, vol. 20, no. 4, pp. 1024-1039, August 2012.

[L25GC2022] Vivek Jain, Hao-Tse Chu, Shixiong Qi, Chia-An Lee, Hung-Cheng Chang, Cheng-Ying Hsieh, K. K. Ramakrishnan, and Jyh-Cheng Chen. 2022. L25GC: a low latency 5G core network based on high-performance NFV platforms. In *Proceedings of the ACM SIGCOMM 2022 Conference*.

[M&M2024] H. Yu, S. Tu, K. K. Ramakrishnan, N. Abu-Ghazaleh, X. Zhang and A. Swami, "M&M: Machine-Learning Enhanced MPQUIC for Resilient Communication with mmWave Networks," *MILCOM 2024 - 2024 IEEE Military Communications Conference (MILCOM)*, Washington, DC, USA, 2024.

[CrowdLoc2018] Ravi Bhandari, Bhaskaran Raman, K. K. Ramakrishnan, Deepthi Chander, Naveen Aggarwal, Divya Bansal, Mahima Choudhary, Nisha Moond, Aneesh Bansal, and Megha Chaudhary. 2018. CrowdLoc: Cellular Fingerprinting for Crowds by Crowds. *ACM Transactions on Sensor Networks* 14, 1, Article 4, February 2018.

☐ Terms:

1) Industrial Technology Classification

Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization